INTERNATIONAL MATHEMATICS CONFERENCE
Geometric Analysis by the Andes
September 7–11, 2026 ▪ Santiago, Chile

INVITED TALK / ABSTRACT

Quantitative stability for the fractional Yamabe problem

Benjamin Borquez

University of California

Abstract

In this talk I will present recent joint work with Sophie Aiken on quantitative stability for the fractional Yamabe problem. The fractional Yamabe problem is a nonlocal analogue of the classical Yamabe problem, in which the conformal Laplacian is replaced by a fractional conformal Laplacian defined through scattering theory on asymptotically hyperbolic manifolds. We study whether functions whose fractional Yamabe energy is close to the minimum must also be quantitatively close to the set of minimizers. We prove that the energy deficit controls the distance to the set of minimizers with a suitable power. The proof combines a Lyapunov–Schmidt reduction for the nonlocal functional with the Caffarelli–Silvestre-type extension characterization of the fractional conformal Laplacian, which allows us to use weighted elliptic PDE techniques. I will also discuss the connection with quantitative stability for the classical and boundary Yamabe problems, as well as some open questions concerning the optimal stability exponent.

Keywords: quantitative stability; fractional Yamabe problem; fractional conformal Laplacian; scattering theory; asymptotically hyperbolic manifolds; fractional Yamabe energy; Lyapunov–Schmidt reduction; weighted elliptic PDE techniques